

PhD student in Neuroimaging and Applied Mathematics.

Education

- Since 2024 **PhD in Neuroimaging**, *Inria Saclay*, Palaiseau, **MIND** team.
Optimal transport applied to fMRI decoding.
- 2023 – 2024 **MSc in Applied Mathematics**, *École Polytechnique*, Palaiseau.
- 2022 – 2023 **MSc in Computer Science**, *Sorbonne Université*, Paris, *Ranked 1/23*.
- 2020 – 2024 **Engineering degree**, *ENSAE Paris*, Palaiseau, specialization in statistics and data science.

Work Experience

- Publications See my [\[Scholar\]](#) page .
- 2024 – 2025 **Causal Inference Instructor**, *Aivancity School for Technology*, Villejuif.
- Oct. 2024 **Instructor at MAIN educational**, *Université de Montréal*.
- 2024 **Master Thesis**, *Inria Saclay*, *MIND* team, Palaiseau, advisor: B. Thirion.
Thesis title: Computing Wasserstein barycenters for efficient inter-subject brain correspondences.
- 2023 – 2024 **Research collaboration**, *Sonio*, Paris, under the supervision of G. Corda.
Implemented data augmentation techniques for echocardiographic images through generative modeling.
- 2023 **Master Thesis**, *Inria Saclay*, *MIND (ex-PARIETAL)* team, Palaiseau, advisors: B. Thirion and S. Dehaene.
Thesis title: Fast Optimal Transport for the Decoding of Human Brains Activity [\[Poster\]](#)
- 2022 – 2023 **Research collaboration**, *Institut du Cerveau – Paris Brain Institute*, Paris, under the supervision of I. Bloch.
Engaged in a research project focused upon spatial analysis of lymphocyte diffusion within a micro-tumor environment, aiming to advance glioma treatment through immunotherapy.
- 2021 **Research internship**, *Pasteur Institute*, Paris, under the supervision of T. Lagache.
Conducted a research internship in a biological image analysis laboratory, resulting in the development of a novel plugin for the ICY software designed to enhance molecular colocalization analysis.

Awards

- 2022 **Winner of the ENSAE Solidarity Hackathon**, *Partnered with Data for Good and Eleven Strategy*.

Languages

- English - C2 level** *Cambridge Linguaskill Certification C1 and above.*
- Spanish - C1 level** *Ease of understanding and communication.*
- French - Native**

Skills

- Languages: Python, C/C++, Zig, R, $\text{\LaTeX}$
- ML frameworks: PyTorch, Jax, Scikit-Learn
- Tools: Git, Github, Docker, Slurm
- Visualization: Matplotlib, Plotly, Shiny

Software

- Nilearn** Machine learning for NeuroImaging in Python – Core Developer.
- fmralign** Functional alignment library for functional Magnetic Resonance Imaging data. – Core Developer.
- FUGW** Scalable Python GPU solvers for Fused Unbalanced Gromov-Wasserstein optimal transport problems – Core Developer.
- BenchOpt** Benchmarking suite for optimization algorithms.